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Protocol

- It is a set of rules or guidelines that defines how two or more devices, systems or programs (computer) communicate with each other.

- Some examples of protocols are:-

① HTTP (Hyper text transfer protocol)

It is a protocol used by the web browser to request and view web pages.

② TCP/IP (Transmission Control / Internet protocol)

It helps computers to communicate over the internet by breaking down information into packets and ensuring that the data reaches the right destination.

- each protocols has its own rules how to send, receive and interpret messages, ensuring that communication is clear and organized.

* How Data is Transferred over the internet

When you send or receive data over the internet data travels through several steps

- ① Request → Imagine you want to visit a website. You open a browser (eg chrome) and type a web address (eg, www.google.com). This is called a request to get data.
- ② Breaking the Request → Your request is broken down into small pieces called data packets. Think of these packets like little envelopes, each with a part of the data and the destination address.
- ③ Sending the Request → The data packets travel across the internet through different devices, cables, routes, and servers, each helping to direct the packet toward its destination.
- ④ Reaching the server → The packets finally reach the server where the website is hosted. The server reads the request and sends back the data. (like web page)
- ⑤ Response → The server sends the data back to your browser, also in the form of small packets. These packets travel back through the internet, following the best route.

- (vi) Reassembly → once all the packets arrive at the browser or computer browser puts the pieces together to show you the full webpage

→ This whole process happens very quickly

* Client-server Architecture

① client -

A client is a device (like computer, phone, tablet) that requests services or data from another device. The client is usually the user end, where people interact with applications

② Server - or system

A server is a powerful computer that provides services, stores data, and responds to requests from clients. Servers run applications, store files, handle user accounts or process data.

- How it Works

- The client sends request to the server asking for data or a service (like asking for website)
- The server processes the request, finds the needed data, and sends it back to the client.

④ Reassembling the Data:-

* Port Numbers

A port number is like a door on a computer or server where specific types of data or services enter or exit.

⊕ How It works

① Identifying services:-

• when you connect to the internet or use a network, your' usually ~~being~~ trying to reach a specific, like a website, email, or chat. each of these services runs on different port

ex→

- port 80 is used for HTTP (web browsing)
- port 443 is used for HTTPS (secure web browsing)
- port 25 is used for email (sending emails)

② Sending Data:-

• when you send a request (eg. loading a webpage), your computer sends the request to specific IP Address (which is like the computer's address), and it specifies the port number (the specific service or application you're trying to reach)

• so, if you type a website address, like "www.example.com", your computer might connect to the IP address of that website and use port 80 to get the web page

(Since websites typically use port 80 for HTTP)

③ How Port Number Work -

- The source port is the port on your computer that's used for sending data (like the "return address" on a letter).
- The destination port is the port on the server or remote computer that is used to handle the specific service you're trying to access (like the "address" where the letter is going).
- Once the data reaches the destination port, the service (like a web server) knows how to process it and send back the correct response.

④ Ports Ranges:-

Port numbers range from 0 to 65535. They are divided into three categories:-

- Well-known ports (0-1023):-

These are reserved for common services like HTTP (port 80), HTTPS (port 443), and FTP (port 21).

- Registered ports (1024-49151).

These are used by specific applications and services (like online games or custom apps).

- Dynamic or private ports (49152-65535)

These are used for temporary connections and are usually assigned by the operating system as needed.

↑

LAN, MAN, WAN

① LAN (Local Area Network)

- A small network covering a limited area, like a home, office or school.
- ex - WiFi in your house connecting your phone, laptop and smart TV.

② MAN (Metropolitan Area Network)

- A bigger network covering a city or town.
- ex - A city's public WiFi or cable TV Network.

③ WAN (Wide Area Network)

- huge network that covers multiple cities or countries.
- ex - the internet itself is the largest WAN.

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Modem, Router

① Modem - "The internet provider"

- A modem brings the internet into your home from your internet service provider (ISP).
- It connects to a cable, phone line or fiber network.
- Without a modem, you can't access the internet.

② Router :- "The Internet Distributor"

- A router distributes the internet from the modem to multiple devices via WiFi or cables.
- It allows multiple devices (phone, laptops, TVs) to connect to the internet at the same time.

* TOPOLOGIES

Topology refers to how devices (computers, routers, switches) are connected in a network.

Main types

① Bus

- All devices are connected to a single central cable (the "bus").
- Data travels in both directions along the cable.

* Pros

- Simple and cheap to setup
- Uses less cable than other topologies

* Cons

- If the main cable fails, the entire network goes down
- More devices slow down the network.

② Star

- All devices connect to a central hub/switch
- The hub/switch manages the network traffic

ex - modern home and office networks using Ethernet or wifi

* Pros

- fast and reliable (each device has its own connection)
- IF one device fails, the network still works

* cons

- IF the central hub fails, everything stops
- Requires more cables, making it costlier than a bus.

③ Ring

- Devices are connected in a circular loop
 - Data moves in one or two directions
- ex - older telecom network and some fiber optic networks.

* Pros

- efficient for small networks
- NO need for a central hub

* cons

- If one device fails, it can break the whole network
- Slower than star topology.

④ Mesh

- every device is connected to every other device
- can be Fully connected (all devices link to all others) or partially connected.

ex - military network and some modern IOT networks.

* Pros

- Highly reliable (if one connection fails, data can take another route)
- Great for large, critical networks.

* cons

- expensive and complicated to set up
- use a lot of cables.

⑤ Tree

- Like a combination of star and bus topologies
- Has multiple levels, with a main central node connecting smaller star networks
- ex - large corporate or university networks.

* Pros

- Scalable
- efficient for large networks.

* cons

- complex setup
- If the main central node fails, a large part of the network can go down

⑥ Hybrid

- A mix of two or more topologies
- used in large, complex networks
- ex - internet service provider.

- The OSI (Open System Interconnection) model is a framework that describes how data moves from one device to another over a network. It has [7] layers each handling a specific task.

* ① Physical layer - The Hardware layer.

Function →

- Transfers raw binary data (0s and 1s) over physical connections (cables, Fiber optics, WiFi)
- Defines network hardware like cables, switches, and radio signals.
- ensures data transmission is electrically or optically correct
- This layer deals with the physical connection between devices

ex - • you're sending a letter, so you need a road or airplane to deliver it
• In a network, this means network cables, WiFi signals, or Fiber optic cables.

Devices at this layer :-

- ethernet cable
- WiFi signals
- Bluetooth
- Fiber optic cables

② Data Link Layer - The Postman layer.

Function →

- This layer organizes data into frames and a MAC address (like a postal address)
- It ensures that data is sent to the correct device within a local network
- It also detects errors in transmission and corrects them

ex -

- A postman delivers your letter to the correct house based on the address.
- In a network, switches use MAC addresses to send data to the right device.

Devices at this layer →

- Network switches
- MAC addresses.

③ Network layer - The post office layer.

Function →

- This layer decides the best router to travel across networks.
- It uses IP addresses (like a home address) to send data from one network to another.
- It breaks data into packets for easier transport.

ex →

- The post office looks at your letter's destination and finds the best delivery route.
- In a network, routers use IP addresses to send data between different networks.

Device at This layer →

- Router
- IP addresses (IPv4, IPv6)

④ Transport layer - The secure parcel layer

Functions →

- This layer ensures complete and reliable data transfer.
- It divides large messages into smaller pieces (segments) and reassembles them at the destination.
- Uses TCP (reliable, ordered data) or

udp (fast, unordered data)

ex→

- A courier company divides a package into multiple boxes and ensures all reach the destination.

- In networking, TCP ensures all packets are received correctly.

* Protocols used Here

- TCP - (transmission control protocol) - used for web browsing, emails (ensures data is ~~complete~~ ^{completely} ^{lately} transferred)

- UDP - (user Datagram protocol) - used for video calls, live streaming (fast but may lose some data)

⑤ session layer - The phone call layer

Function

- Manages the start, maintenance, and end of communication (session) between devices.

- ensures a session remains active while communication happens.

ex →

- you call a friend before sending a parcel to ensure they are home.
- In networking, this layer establishes a connection before transferring data.

* Where It's used

- Video conferencing (Zoom, Google Meet)
- Logging into websites (session tokens)

⑥ Presentation layer - the translator layer

Functions →

- Converts data into a format that both sender and receiver understand.
- Handles encryption and compression (reducing file size)

ex →

- If you send a letter in different language, this layer translates it before delivery
- In networking, it encrypts web data using SSL/TLS to secure it.

Where it's used

- SSL/TLS encryption (HTTPS websites)
- JPEG, MP3, MP4 formats.

⑦ Application layer - The user interaction layer.

Function →

- The layer that users interact with (websites, apps, etc).
- It allows users to send and receive data over the network

ex →

- The receiver opens the letter and reads it
- In networking, this layer handles web browsers, emails, and file transfers.

- web browsers (chrome, Firefox, edge)
- Email apps (Gmail, outlook)
- File transfer (FTP, Google Drive)

TCP/IP

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- The TCP/IP model (transmission control protocol/ internet protocol) is a simplified version of the OSI model that explain how data moves across the internet. It has 4 layers
- It is Faster and simpler than the OSI model.

① Network Access layer (Link layer)

Function

This layer is responsible for how data is physically sent over a network (like WiFi, Ethernet, Fiber)

- It deals with MAC addresses, physical cables, and network interfaces
- Ensures data is correctly formatted for transmission over different types of networks

ex

Think of this as the road on which cars (data) travels. Different roads (WiFi, Ethernet) have different rules of driving

② Internet layer (Network layer)

- This layer helps route data between different networks using IP addresses
- The most important protocol here is IP which assigns unique addresses to devices
- It ensures that data packets reach the correct destination even if they pass through multiple networks.

ex

Imagine sending a letter with an address on it. No matter how many post offices (routers) it goes through it eventually reaches the right destination.

③ Transport layer.

- This layer ensures reliable data transfer between devices
- The two main protocols here are:-
 - TCP - (Transmission Control Protocol) - Ensures data arrives completely and in order (like phone call)
 - UDP - (User Datagram Protocol) - Faster but does not guarantee complete delivery (like a live video stream)

ex - TCP is like Formal conversation where you confirm each sentence before continuing.
UDP is like shouting information at someone without waiting for confirmation.

④ Application layer.

- This is where user application like web browser, emails and messaging apps operate.

- common protocols include:-

- HTTPS/HTTP (For Websites)

- FTP (For File transfers)

- SMTP/IMAP (For emails)

ex →

When you visit website, your browser uses HTTP to request a web page from a server.

* How Data Moves Through the layers

- ① A user sends a message (Application layer).
- ② The message is broken into packets (transport layer).
- ③ each packet gets an IP address (Internet layer).
- ④ The packets are physically sent over the network (Network Access layer).
- ⑤ At the destination, the layers work in reverse to reconstruct the message.

* Why is the TCP/IP Model important?

- It allows different types of devices to communicate.
- It is the foundation of the internet.
- It provides a standardized way of sending and receiving data.

Comparison

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OSI Model (7 layers)	TCP/IP Model (4 layers)	Explanation
1. physical layer	(part of Network Access layer)	Deals with cables, switches, and physical connections.
2. Data link layer	(part of Network Access layer)	Handles MAC addresses and error detection.
3. Network layer	internet layer	Responsible for routing packets using IP addresses.
4. transport layer	transport layer	Ensures reliable (TCP) or Fast (UDP) communication
5. Session layer	(part of Application L)	Manages sessions between devices (e.g. keeping you logged into a website)
6. presentation layer	(part of Application L)	Formats data (e.g. encryption, compression)
7. Application layer	Application layer	Handles user applications like browsers, emails and messaging

Peer to peer Architecture

- * peer to peer architecture is a type of network design where devices (called peers) communicate directly with each other without relying on a central server. Each device in the network acts as both a client and a server,
- * meaning it can request and provide services to other devices.

→ How Does P2P work?

- ① each peer (computer or device) is equal in status - there is no central authority
- ② peers can share resources (files, processing power or internet bandwidth)
- ③ when a peer needs data or service, it can connect directly to another peer that has it
- ④ The system grows as more peers join, distributing the workload.

Examples →

- ① File sharing (BitTorrent, uTorrent)
 - when you download a movie or software using a torrent, you're not getting it from a single server.
 - instead, your computer downloads pieces of the file from multiple peers who already have parts of it
 - At the same time, you are also sharing the parts you downloaded with others.

② cryptocurrency (Bitcoin, Ethereum)

- There is no central bank or authority managing transactions.

- Every participant (node) in the network validates and records transactions on a shared ledger (blockchain)

③ Gaming (LAN multiplayer Games, some online Games)

- Some multiplayer games allow players to connect directly to each others computers to play, rather than relying on game servers.

* Types of P2P Networks

① Pure P2P

- NO central server or authority
- All peers are Fully equal
- Ex - early versions of Napster (music sharing services)

② Hybrid P2P

- use a central server to help find peers but does not store the actual data.
- once peers connect, they communicate directly
- Ex - modern BitTorrent clients use this method.

③ Structured P2P

- peers are organized in a structured way like a distributed database
- uses algorithms to efficiently search for and locate data.
- Ex - Distributed Hash tables (DHT) in BitTorrent)

* Advantages

- 1) NO central Server Required.
- 2) Scalability
- 3) cost - Effective
- 4) Efficient Resource utilization.

* Disadvantages

- 1) Security Risks
- 2) Data Integrity Issues
- 3) Slower Search Speed
- 4) Bandwidth consumption.

Types of protocols

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- HTTP (HyperText Transfer protocol) :-
Rules For browsing the web - like how your browser asks for and gets website.

- TCP/IP (Transmission control protocol / internet protocol)

The basic rules for sending and receiving data over the internet, making sure it arrives in order and complete.

- FTP (File Transfer Protocol) :-

Rules For uploading or downloading files between devices.

Mostly we don't use this now because HTTP also support this.

- UDP (User Datagram protocol)

A fast way to send data (like for video streaming or gaming where speed matters more than perfect order).

- DNS (Domain Name system)

Rule translating website name (like google.com) into number (IP addresses) devices understand.

- SMTP (Simple mail transfer protocol)

Rules for sending mail

- POP/IMAP (Post Office Protocol / Internet Message Access Protocol)

– Rules for receiving emails – POP downloads them, IMAP syncs them

- ARP (Address Resolution Protocol)

Rules for finding a device's hardware address (MAC) using its network address (IP)

Socket

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- * Think of a Socket as a "connection point" or a "door" that allows two devices (like computers or programs) to talk to each others over a network such as the internet.

It's a way for software to send and receive data, kind of like how a phone line connects two people for a conversation.

How Does a Socket Work

Sockets are used when devices or programs need to communicate.

- ① Two sides:- This usually a "server" and a "client" For example, when you visit a website.

* How Does It Work?

When two computers want to chat they need two key pieces of info:-

- ① IP Address

- ② port Number

A socket combines these two things (IP+port) to make a unique endpoint for ex:-

computer A might have a socket at

192.168.1.1 : 5000 (

IP port

computer B might have a socket at

172.16.0.2 : 3000

These two sockets can connect, and then data - like text, image, or videos - can flow between them

+ Types of Socket

① Stream Socket (TCP):-

- This is like a phone call: reliable, orderly, and two-way

- Data arrives in the exact order it was sent, and nothing gets lost.

- used for things like loading a webpage or sending an email

ex:- visit google.com and a stream socket makes sure all the webpage data arrives correctly.

② Datagram Socket (UDP):-

- This is more like mailing postcards: fast but less reliable.

- Data might arrive out of order, or some might not show up at all.

- used for things like live video streaming or online gaming, where speed matters more than perfection.

HTTP

(Application
layer
protocol)

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- * HTTP stands for Hyper Text Transfer Protocol. It's the system that allows your web browser (like chrome or Firefox) to request stuff - like a webpage - from a server (a computer that stores websites) and get it delivered back to you. It is the foundation of how we use the internet to browse website.

* How Does it Work?

① You ask for something (Request): when you type a website address (like www.example.com) into your browser and hit enter, your browser sends an HTTP request to the server hosting that website. This request says, "Hey, can you send me this webpage?"

② The server Responds:- The server gets your request, prepares the webpage (or whatever you asked for), and sends it back with an HTTP response. your browser then shows it to you.

* Key parts of HTTP-

- client and server:- The client is your device making the request. The server is the computer that has the stuff you want to send it back.

- Messages :- HTTP uses two main types of messages.

- Request
- Response.

- Methods :- These are like instructions in the request common ones are:-

- ① GET :- Please give me this thing (like loading a webpage).

- ② POST :- Here is some info, save it (like submitting a form).

- ③ PUT :- update this thing (like editing a profile).

- ④ DELETE :- Remove this thing (like deleting a post).

- Status codes :-

HTTP status code indicates that the result of the request

1. 1xx - Informational

- 100 continue - The server received the initial request, continue with the rest

- 101 switching protocols -

- Switching protocols as requested.

2. 2xx - Success

- 200 - OK - Request Succeeded

- 201 - created - New resource created Successfully

- 204 - NO content - NO content to return

3. 3xx - Redirection

- 301 Moved Permanently - Resource moved to a new URL

- 302 Found - Resource temporarily moved

- 304 Not Modified - Resource not modified since the last request.

4. 4xx - client Errors -

- 400 Bad Request - Malformed request

- 401 unauthorized - Authentication required.

- 403 Forbidden - Access denied

- 404 Not found - Resource not found.

6. 5XX - Server Errors

- 500 Internal Server Error - Generic Server error
- 502 Bad Gateway - Invalid response from upstream Server.
- 503 service unavailable - Server is unavailable.

• cookies and sessions in HTTP -

• cookies:-

Small pieces of data stored on the client to maintain Session state.

• session:-

Server-side mechanism to store user data between request

* cookies - How cookies work

① Server Sends cookie.

- when a user logs in or performs an action, the server sends a cookie to the client
- The response includes a set-cookie header:-

ex:-

HTTP/1.1 200 OK

Set-cookie: sessionId = abc123;

path = /;

HttpOnly

② client stores cookie:-

The browser stores the cookie and sends it with subsequent requests.

ex:-

GET /dashboard HTTP/1.1

cookie: sessionId = abc123

③ server receives cookie:-

The server reads the cookie and validates the session or identifies the user.

• common cookie Attributes

1. Name and value

• The key-value pair identifying the cookie

ex -

set-cookie: user = JohnDoe;

2. expires / max-age

• specifies when the cookie should

expire

ex -

Set-cookie: sessionId = abc123;

Expires = Wed, 21 Oct 2025 ...

3. path

Limits the cookie to a specific path on domain

ex

Set-cookie: path = /dashboard

4. Domain

- Specifies the domain where the cookie is valid.

ex - Set-cookie: Domain = exam.com

5. Secure

- Ensures the cookie is only sent over HTTPS.

6. HTTPOnly

- prevents JavaScript from accessing the cookie, mitigating XSS attacks.

7. SameSite

- controls whether cookies are sent with cross-site requests.

• use of cookies

- Authentication: maintain logged-in state.
- user preference: store theme, language, etc.
- Tracking: Track user behaviour for analytics.

★ Sessions

A session is a server-side mechanism that stores user information across multiple requests.

Sessions are typically used to maintain user authentication state and application-specific data.

★ How Sessions work

- ① client sends login Request
 - user logs in, and the server creates a session.
 - The server generates a unique sessionId and stores session data.

- ② server sends sessionId as a cookie.

ex - set-cookie: sessionId= abc123;
HttpOnly;
secure

③ client stores and sends session ID.

- The browser stores the session ID in a cookie.
- The client sends this cookie in subsequent requests.

④ server validates session ID

- The server retrieves the session data using the session ID.
- IF the session is valid, the user gets access.

* Session storage options

① In-memory storage

- Fast but not persistent.
- Loses data if the server restarts.

② Database storage

stores session data in a database like MongoDB, Redis, or SQL.

③ File system storage.

- Stores session data in files.
- Suitable for small applications.

Session Management Techniques

① session ID in cookie.

- most common approach
- session ID stored in an HTTP-only secure cookie.

② Token-Based session (JWT)

- stores session data in a self-contained token (JWT)
- No need for server-side session storage.

How Email Works

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① composing and sending an Email

When you write an email in an email client such as Gmail, Outlook you enter:-

- Recipient's Email Address :- who you're sending to
- Sender's Email Address (your email)
- Subject and message content
- Attachments (if any)

② sending to the outgoing mail server (SMTP Server)

Your email client sends the message to an SMTP (Simple Mail Transfer Protocol) Server, which acts like a post office.

- SMTP Server checks the recipient's domain (e.g. @gmail.com, @yahoo.com)
- It looks up where to send the email using DNS to find the recipient's mail Exchange (MX) server.

ex:- if I am sending an email to user@yahoo.com, my Gmail SMTP Server will look up Yahoo's mail server and forward the email there.

③ Delivering the Email to the Recipient's mail server.

Once the SMTP server finds the correct mail server (mail Exchange - Mx) for the recipient, it delivers the email to that server.

- The recipient's mail server stores that email in their inbox (Mailbox server)

- This mail server uses IMAP (Internet message Access protocol) or pop3 (post office protocol) to allow access.

* **IMAP** - Internet Message Access protocol

- emails stay on the server.

(For

Gmail is Google's mail server

Yahoo is Yahoo's mail server)

- When you open your email client it fetches emails without downloading them permanently

- You can access your emails from multiple devices because the emails remain stored on the server.

- If you delete an email from one device, it gets deleted from all device the changes are synced with the server

*

POP3 - Post Office Protocol 3

- Email are downloaded to your device and then removed from the server (unless configured to keep a copy)
- you can read your emails even without an internet connection because they are stored locally
- if you access email from another device later, it won't show older emails because they were already downloaded to the first device.

④ Receiving the Email

When the recipient opens their email app, their device connects to their mail server using

① IMAP

② POP3

⑤ Reading and Replying

The recipient can now read, reply, forward, or delete the email

* Protocols involve in email communication

① SMTP ② IMAP ③ POP3

④ DNS.

DNS

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- DNS (Domain Name System) is like the phonebook of the internet. It translates human-readable domain name like google.com into IP addresses like 142.250.195.78, which computers use to identify each other on the network.

Without DNS you'd have to remember IP addresses to visit websites - which is obviously not user-friendly.

* How DNS Works

- ① You type www.example.com in your Browser.
 - Your browser checks if it has this domain's IP cached.
 - If not, it asks your operating system to resolve the domain.
- ② OS checks local DNS cache
 - Your computer might have recently resolved the same domain, so it checks its local cache.
 - If it finds nothing, it asks the configured DNS Resolver (usually provided by your ISP or services like Google DNS 8.8.8.8 or ~~cloud~~ Cloudflare DNS 1.1.1.1).

③ DNS Recursive Resolver (Resolver)

a. contacts Root DNS Server

- It ~~he~~ asks one of the 13 root servers: "Where can I find .com TLD info?"

b. Root Server Responds

- points the resolver to TLD (top level domain) server for .com

c. TLD Server Responds

- points to the Authoritative Name server for example.com.

d. Authoritative Name server.

- This server has the actual IP address (A record) for www.example.com

- Resolver gets this IP and returns it to your browser.

④ Browser. connects to the IP.

- your browser sends HTTP/HTTPS request to it

* caching in DNS

DNS responses are cached to improve speed and reduce load

- Browser cache
- OS cache
- Resolver cache
- Authoritative cache

* Public DNS providers

- Google DNS
- CloudFlare
- Open DNS

Deep Dive Into Transport Layer.

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* What is the Transport layer?

The Transport layer is one of the layers in the networking model like OSI and TCP/IP that handles the delivery of data between two devices.

→ Simple way to think about it

Imagine you're sending a package (data) to a friend far away. The transport layer is like a delivery company that makes sure the package:

- is sent correctly
- Reaches the correct friend
- is complete, not broken
- can be tracked and managed.

* What does the Transport layer do?

1. Segmentation and Reassembly

- It breaks big data into smaller parts so they can travel easily over the network
- later it reassembles these parts back into the full data at the receiving end.

2. End-to-End communication.

- It creates a direct connection between the sender and receiver
- It manages the conversation, like making sure both sides understand

each other.

3. Flow control

- It controls the speed of data sending so that the receiver isn't overwhelmed.

4. Error control

- It detects if any data is missing or broken
- If something is wrong, it can ask for a resend.

5. Multiplexing and Demultiplexing

- It can handle multiple conversations at the same time.

ex - on computer, we might be downloading a file and having a video call - Transport layer manages both separately

* so in simple words.

- It connects your device directly to another.
- It breaks data into small pieces and rebuilds it on the other side.
- It checks for mistakes and controls speed.
- It handles multiple conversations at once.

checksum In Transport layer.

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- A checksum is a small value (a number) that is calculated from the original data.
- It is like a digital fingerprint of the data.

* Why do we use checksum?

- To check if the data was changed, lost, or damaged during transmission.
- If the checksum doesn't match at the receiver's side, it means something went wrong with the data.

* How does a checksum work?

1) The sender calculates a checksum from the data and sends both the data and checksum (done by transport layer)

2) The receiver receives the data and calculates the checksum again.

3) Then the receiver compares the two checksums:-

- If they are the same, the data is correct

- If they are different, the data is corrupted.

Timers In Transport layer.

- Timers in the Transport layer are used to track the timing of actions and ensure data is delivered properly.

* Why do we use Timers?

- to manage retransmissions: If data isn't acknowledged by the receiver within a certain time frame, a timer will trigger a retransmission of the lost or delayed packet.

- to prevent waiting forever:- if the data doesn't arrive within a set time the sender knows something went wrong and takes action (e.g. - resend the data, alert, etc)

* How do Timers work?

1. When the sender sends data (a segment), it starts a timer.
2. The sender waits for an acknowledgment from the receiver.
3. if the Ack comes back before the timer expires, the sender moves on to the next segment.
4. if the timer expires and no Ack received, the sender resends the segment.

UDP (User Datagram protocol) is one of the main protocols in the transport layer of the Internet protocol suite.

It is used to send data quickly over a network without guaranteeing

* Why UDP?

Sometimes, speed is more important than accuracy. For example:

- online games
- video streaming
- voice calls
- live broadcasts.

These applications can tolerate some data loss but they can't tolerate delays.

* How UDP works

1. connectionless

UDP does not establish a connection before sending data. It just sends.

2. No handshakes

Unlike TCP (which does 3-way handshake) UDP just puts data in a packet and sends it out.

3. No guarantees

- That the data reaches the receiver
- That packets arrive in the right order
- That packets are not duplicated

4. Lightweight

UDP headers are small (only 8 bytes) which makes it fast and efficient.

* UDP packet structure (Header)

Field	Size	Description
Source port	2B	sender's port no
Destination port	2B	Receiver's port no
length	2B	Total length (header + data)
checksum	2B	For basic error detection.

Deep Dive Into TCP

classmate

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TCP (Transmission control protocol) is a protocol in the transport layer used to reliably send and receive data over a network.

It is used when accuracy and order matter more than speed - like sending a message, a file, or loading a web page.

* Why TCP?

- You must not lose any data.
- You need data to arrive in the right order.
- You want confirmation that data was delivered.

* How TCP Works (Simply):-

1. connection-oriented

- TCP establishes a connection first before sending data.
- Uses a 3-way handshake to start communication.

2. Reliable Delivery

- It makes sure every packet is received.
- If something gets lost or corrupted TCP resends it.

3. ordered Delivery

- Packets may arrive out of order but TCP reorders them.

4. Flow control

Ensures the Sender doesn't send too Fast For the receiver to handle.

5. congestion control

- Avoids overloading the networks

* congestion control

congestion control is a network mechanism that helps prevent too much data from being sent through the network at once, which can overload routers or links and cause delays, packet loss, or even total failure of communication.

* TCP structure (packet)

Field

Source/Dest port

sequence number → order of packets

Acknowledgment → confirms received data.

Flags (SYN, ACK) → controls connection & Flow

checksum → error checking

TCP VS UDP

Feature	TCP	UDP
→ connection	yes	NO
→ speed	Slower	Faster
→ order & Delivery	Guaranteed	Not guaranteed
→ congestion control	yes	NO
→ use case	web, files emails	Games, live video, voice.

3-way Handshake

The 3-way handshake is the process that TCP uses to establish a connection between two devices (like your phone and a server) before they start exchanging data.

* Steps of the TCP 3-way Handshake

1. SYN (Synchronize) - "Hello, can we talk?"

- client → server
- The client sends a SYN packet saying:-
 - I want to start a connection
 - Here's my initial sequence number.

2. SYN-Ack (Synchronize + Acknowledge) - "yes and here's my info!"

- server → client

- The server responds with a SYN-Ack packet

∴ I accept your request

∴ Here's my own sequence no

∴ I acknowledge your sequence number.

3. ACK (Acknowledge) - "Great, let's go!"

- client → server

- The client send an Ack packet:

- "Got your message, were good to go!"

* Why is the 3-way handshake Important?

- ensures both sides are ready
- verifies network paths work
- Synchronizes sequence numbers

Network Layer

Deep Dive.

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* In networking, data travels from one computer to another through different layers, like steps in a process. These layers are defined in the OSI model which has 7 layers. The Network layer is layer 3.

* Main purpose of the Network layer.

The network layer is responsible for

1. Routing → Figuring out the best path for data to travel from sender to receiver
2. Logical addressing → using IP addresses to identify the sender and receiver
3. Packet forwarding → sending data packets from one network to another.

* How it works

• like we are sending a message from your laptop to your ~~far~~ Friend's laptop who lives in another city

• This message goes as data packets through routers and switches

- ① your data gets broken into packets
- ② each packet is given a source IP and a destination IP
- ③ The network layer looks at the destination IP and decides which path the packet should take (using a routing table)
- ④ It sends the packet to the next router on the way

⑤ each router it passes through does the same reads the destination IP and forwards it along

⑥ eventually, the packet reaches your friend's laptop

* Key Terms you should know

IP Address → Logical address of a device

Packet → A small chunk of data sent over the network

Router → A device that forwards packets between networks

Routing table → A table routers use to know where to send packets.

IPv4/IPv6 → versions of IP addresses

* What's inside a packet?

IP Header :- Includes source IP, destination IP, TTL (time to live), etc

payload :- Actual data being sent

The TTL helps prevent packets from circulating forever. Every time the packet passes a router, TTL is reduced by 1. If it hits 0, the packet is dropped.

Control plane

→ The control plane is the brain of a network device (like a router or switch).

→ It's responsible for making decisions about where traffic should go (packets)

like:-

- Building routing tables
- choosing the best path
- exchanging routing information with other devices.

Routing table

A routing table is a list of rules inside a router that tells it where to send packets based on their destination IP address.

Think of it like a Google maps for routers, it tells the router.

"If you want to go here (destination network), take this path (next hop)."

* What's inside a Routing table?

Field	Meaning
Destination	→ The IP address range the packet is going to
Subnet mask	→ How much of the destination address to match.
Next Hop	→ The next router (IP address) to forward the packet to
Interface	→ Which router port to send it out from
Metric/cost	→ The preference for the route (lower is better)

* How the Router uses the Routing table when a packet arrives:-

1. The router looks at the destination IP in the packet
2. It searches the routing table to find the best match.
3. It forwards the packet out through the correct interface to the next hop

Internet protocol

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Internet protocol is a set of rules that governs how data is sent and received over the internet. It ensures that data packets travel from one computer to another correctly.

* key concepts of Internet protocol

① IP Address

- A unique identifier for a device on a network

- Example : 192.168.1.5 (IPv4) or (IPv6) to complex

It helps identify who is sending and who should receive the data.

② IP Packet

- The data unit that travels across the internet

- It contains

- Header (source IP, destination IP)

- payload (actual data)

③ Routing

- IP helps find the best path to send packets from source to destination

- Routers read the destination IP and forward the packet to the next best hop.

IP Address

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— An IP address is a unique identifier assigned to each device connected to a network. It works just like your home address, helping devices find and communicate with each other over the internet.

* Why Do We Need IP Addresses?

Imagine we have to send the letter

- you need the receiver's address
- you include your address so they can reply

Similarly, devices use IP addresses to:-

- send data to the correct device
- Receive responses from that

device

* Two versions of IP Addresses:-

① IPv4

② IPv6

① IPv4 -

Format → ① 32 bit address

② written as 4 decimal numbers

separated by dots

ex - 192.168.1.1

each number can range from 0 to 255

How Many Addresses?

- Around 43 billion unique addresses
- With the rapid growth of the internet (phones, IOT, etc), IPv4 addresses are running out

② IPv6

- IPv4 couldn't handle the growing number of devices
- IPv6 provides a massive number of addresses - about 340 undecillion (3.4×10^{38})

Format:-

- 128-bit address
- written as 8 group of 4 hexadecimal digits, separated by colons.
- ex -

2001:0db8:85a3:0000:0000:8a2e:0370:7334

Middle boxes

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* A middlebox is any network device between the sender and receiver that performs Functions other than standard IP Forwarding

In simple terms:-

- A router forwards packets from source to destination

- A middlebox modifies, filters, inspects, or manages those packets for security, optimization, or control

→ example

① Firewall

② NAT

③ Proxy Server

④ Load Balancer

⑤ WAN

* Why we use

① Security

② Performance

③ Policy enforcement

④ Address Management.

Data Link Layer

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- The data link layer is layer 2 in the OSI model
- It sits just above the physical layer (layer 1) and just below the network layer (layer 3)
- Its main role is to ensure error-free transmission of data between two devices on the same local network.

* Key Responsibilities of the Data Link Layer

Framing → Divides the data into frames for easier transmission.

Error Detection and Correction → Adds mechanisms like CRCs to detect and sometimes correct errors during transmission.

Flow Control → Ensures the sender doesn't overwhelm the receiver.

MAC Addressing → Uses physical (MAC) addresses to identify devices on the same network.

* How It Works

sending a File between two PCs on the same LAN.

- ① Application layer breaks the file into data
- ② Transport & Network layers wrap it with headers
- ③ Data link layer:-
 - Divides data into frames
 - Adds MAC addresses
 - Adds error detection bits
- ④ physical layer converts frames to electrical signals or light
- ⑤ Receiver's Data link layer:-
 - checks for error
 - confirms the MAC address
 - passes the clean data up to layer 3.

* common technologies in data link layer:-

- ethernet
- Wi-Fi
- PPP
- HDLC
- Frame Relay